

Supplementary Materials

Reading skill and the time course of morphological processing: electrophysiological evidence for stage-specific modulation

Overview

This document reports full model output for analyses summarised briefly in the main text. Sections S1–S2 report the word morphological family size analyses. In the main manuscript, base frequency is treated as the primary morphological predictor for word ERP analyses, since the study’s central claim — that Orthographic Sensitivity (OS) and Language Proficiency (LP) show distinct, component-specific patterns of modulation, with OS the primary modulator at the N250 and LP the primary modulator at the N400 — is formally tested using base frequency as the morphological variable of interest. Family size analyses for words were conducted in parallel, using the same model structure, and are summarised briefly in the main text (see *Words N250: Base Frequency* and *Words N400: Base Frequency*). Sections S3–S4 report exploratory models that reverse the interaction structure at each time window, testing whether the non-predicted individual difference dimension — Language Proficiency at the N250, Orthographic Sensitivity at the N400 — also modulated processing of base frequency and family size, with the predicted dimension included as a covariate. The full statistical output for all four exploratory models, along with the two family size models, is reported below for transparency and to allow readers to evaluate each pattern in full.

S1. Words N250: Family Size

Mean N250 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Family Size, Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Family Size and OS. The model included random intercepts for participants and for electrode site nested within participant. The analysis revealed a robust main effect of Family Size—words from smaller morphological families elicited more negative N250 amplitudes than words from larger families, $M_{\text{large}} = -0.57 \mu\text{V}$, 95% CI $[-1.03, -0.11]$, $M_{\text{small}} = -0.90 \mu\text{V}$, 95% CI $[-1.36, -0.44]$, $\beta = 0.324$, $SE = 0.058$, $t(1618) = 5.54$, $p < .001$, $\eta_p^2 = .02$. Neither Orthographic Sensitivity, $\beta = 0.261$, $SE = 0.256$, $t(57) = 1.02$, $p = .31$, nor Language Proficiency, $\beta = -0.279$, $SE = 0.187$, $t(57) = -1.50$, $p = .14$, yielded reliable main effects. The interaction between Family Size and Orthographic Sensitivity did not reach significance but showed a marginal trend, $\beta = -0.111$, $SE = 0.065$, $t(1618) = -1.70$, $p = .089$, $\eta_p^2 = .002$, suggesting that the family

size effect at the N250 may be modulated by orthographic skill, though this requires cautious interpretation given the marginal status of the effect. The model explained 1.9% of the variance in N250 amplitude via fixed effects ($R^2_{\text{marginal}} = .019$) and 73.8% overall ($R^2_{\text{conditional}} = .738$), with the large conditional R^2 reflecting substantial variability in N250 amplitude across participants and electrode sites. Together, these results indicate that morphological family structure robustly modulates early morpho-orthographic processing for words, with limited but suggestive evidence that orthographic skill may shape the sensitivity of this effect.

Table 1: Words N250: Family Size model, fixed effects.

Term	β	SE	df	t	p
Family Size	0.324	0.058	1618	5.54	< .001
Orthographic Sensitivity	0.261	0.256	57	1.02	.31
Language Proficiency	-0.279	0.187	57	-1.50	.14
Family Size $\times$ OS	-0.111	0.065	1618	-1.70	.089

$R^2_{\text{marginal}} = .019$; $R^2_{\text{conditional}} = .738$. Random effects: participant, electrode nested within participant.

S2. Words N400: Family Size

Mean N400 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Family Size, Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Family Size and LP. The model included random intercepts for participants and for electrode site nested within participant. The analysis revealed a robust main effect of Family Size—words from smaller morphological families elicited greater N400 amplitudes than words from larger families, $M_{\text{large}} = 0.518 \mu\text{V}$, 95% CI $[-0.054, 1.089]$, $M_{\text{small}} = 0.062 \mu\text{V}$, 95% CI $[-0.509, 0.634]$, $\beta = 0.452$, $SE = 0.064$, $t(1618) = 7.07$, $p < .001$, $\eta_p^2 = .03$, indicating facilitated semantic integration for words supported by richer morphological families. A significant main effect of Orthographic Sensitivity was also observed, $\beta = 0.677$, $SE = 0.318$, $t(57) = 2.13$, $p = .038$, $\eta_p^2 = .07$, reflecting overall differences in N400 amplitude as a function of orthographic skill independently of morphological structure. Language Proficiency did not yield a reliable main effect, $\beta = 0.001$, $SE = 0.231$, $t(57) = 0.005$, $p = .996$, nor did the Family Size $\times$ LP interaction reach significance, $\beta = -0.071$, $SE = 0.052$, $t(1618) = -1.37$, $p = .172$. These results indicate that semantic integration of morphological family information for familiar words is robust and not selectively modulated by individual differences in language proficiency.

Table 2: Words N400: Family Size model, fixed effects.

Term	β	SE	df	t	p
Family Size	0.452	0.064	1618	7.07	< .001
Language Proficiency	0.001	0.231	57	0.005	.996
Orthographic Sensitivity	0.677	0.318	57	2.13	.038
Family Size $\times$ LP	-0.071	0.052	1618	-1.37	.172

S3. Words N250: Exploratory Language Proficiency Models

To assess whether Language Proficiency also modulated the N250 base frequency effect for words, an exploratory model was fit with fixed effects of Base Frequency, Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Base Frequency and LP; this reverses the structure of the primary model reported in the main text, in which OS rather than LP was the interacting term. The model included random intercepts for participants and for electrode site nested within participant. The main effect of Base Frequency was significant, $\beta = -0.329$, $SE = 0.100$, $t(1618) = -3.27$, $p = .001$, $\eta_p^2 = .007$, consistent with the primary model. Neither Language Proficiency, $\beta = -0.536$, $SE = 0.378$, $t(57) = -1.42$, $p = .162$, nor Orthographic Sensitivity, $\beta = 0.534$, $SE = 0.520$, $t(57) = 1.03$, $p = .309$, yielded reliable main effects. Critically, the interaction between Base Frequency and Language Proficiency was significant, $\beta = 0.180$, $SE = 0.082$, $t(1618) = 2.20$, $p = .028$, $\eta_p^2 = .003$. As reported in the main text, follow-up contrasts indicated that this modulation ran in the opposite direction to the Orthographic Sensitivity interaction in the primary model: the base frequency effect was largest at low LP and attenuated steadily as LP increased, becoming non-significant by LP = +1 SD ($\Delta = -0.15 \mu V$, $t(1618) = -1.13$, $p = .259$) and reversing in direction, though not significantly, at the upper end of the observed range (LP = +3 SD: $\Delta = 0.21 \mu V$, $t(1618) = 0.78$, $p = .433$). The model explained 1.8% of variance via fixed effects ($R_{\text{marginal}}^2 = .018$) and 79.9% overall ($R_{\text{conditional}}^2 = .799$).

Table 3: Words N250, Base Frequency $\times$ LP (exploratory), fixed effects.

Term	β	SE	df	t	p
Base Frequency	-0.329	0.100	1618	-3.27	.001
Language Proficiency	-0.536	0.378	57	-1.42	.162
Orthographic Sensitivity	0.534	0.520	57	1.03	.309
Base Frequency $\times$ LP	0.180	0.082	1618	2.20	.028

$R_{\text{marginal}}^2 = .018$; $R_{\text{conditional}}^2 = .799$. Random effects: participant, electrode nested within participant.

A parallel exploratory model tested the interaction between Family Size and Language Proficiency at the N250, with fixed effects of Family Size, Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Family Size and LP; Orthographic Sensitivity was again included as a covariate. The main effect of Family Size was significant, $\beta = 0.325$, $SE = 0.058$, $t(1618) = 5.56$, $p < .001$, $\eta_p^2 = .02$, consistent with the primary family size model reported in Table S1. Neither Language Proficiency, $\beta = -0.279$, $SE = 0.186$, $t(57) = -1.50$, $p = .140$, nor Orthographic Sensitivity, $\beta = 0.261$, $SE = 0.256$, $t(57) = 1.02$, $p = .313$, yielded reliable main effects. The interaction between Family Size and Language Proficiency was not significant, $\beta = -0.074$, $SE = 0.047$, $t(1618) = -1.56$, $p = .119$, indicating that, unlike base frequency, the family size effect at the N250 was not modulated by Language Proficiency. The model explained 1.9% of variance via fixed effects ($R_{\text{marginal}}^2 = .019$) and 73.8% overall ($R_{\text{conditional}}^2 = .738$).

Table 4: Words N250, Family Size $\times$ LP (exploratory), fixed effects.

Term	β	SE	df	t	p
Family Size	0.325	0.058	1618	5.56	< .001
Language Proficiency	-0.279	0.186	57	-1.50	.140
Orthographic Sensitivity	0.261	0.256	57	1.02	.313
Family Size $\times$ LP	-0.074	0.047	1618	-1.56	.119

$R^2_{\text{marginal}} = .019$; $R^2_{\text{conditional}} = .738$. Random effects: participant, electrode nested within participant.

S4. Words N400: Exploratory Orthographic Sensitivity Models

To assess whether Orthographic Sensitivity also modulated the N400 base frequency effect for words, an exploratory model was fit with fixed effects of Base Frequency, Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Base Frequency and OS; this reverses the structure of the primary model reported in the main text, in which LP rather than OS was the interacting term. The model included random intercepts for participants and for electrode site nested within participant. The main effect of Base Frequency showed a marginal trend, $\beta = 0.189$, $SE = 0.107$, $t(1618) = 1.77$, $p = .077$, $\eta_p^2 = .002$. Orthographic Sensitivity showed a significant main effect, $\beta = 1.290$, $SE = 0.640$, $t(57) = 2.02$, $p = .049$, $\eta_p^2 = .07$, consistent with the OS main effect reported in the primary model. Language Proficiency did not yield a reliable main effect, $\beta = 0.047$, $SE = 0.465$, $t(57) = 0.10$, $p = .920$. The interaction between Base Frequency and Orthographic Sensitivity was not significant, $\beta = 0.195$, $SE = 0.120$, $t(1618) = 1.63$, $p = .104$, indicating that Orthographic Sensitivity did not modulate the N400 base frequency effect. The model explained 2.6% of variance via fixed effects ($R^2_{\text{marginal}} = .026$) and 82.4% overall ($R^2_{\text{conditional}} = .824$).

Table 5: Words N400, Base Frequency $\times$ OS (exploratory), fixed effects.

Term	β	SE	df	t	p
Base Frequency	0.189	0.107	1618	1.77	.077
Orthographic Sensitivity	1.290	0.640	57	2.02	.049
Language Proficiency	0.047	0.465	57	0.10	.920
Base Frequency $\times$ OS	0.195	0.120	1618	1.63	.104

$R^2_{\text{marginal}} = .026$; $R^2_{\text{conditional}} = .824$. Random effects: participant, electrode nested within participant.

A parallel exploratory model tested the interaction between Family Size and Orthographic Sensitivity at the N400, with fixed effects of Family Size, Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Family Size and OS. The main effect of Family Size was significant, $\beta = 0.452$, $SE = 0.064$, $t(1618) = 7.07$, $p < .001$, $\eta_p^2 = .03$, consistent with the primary family size model reported in Table S2. Orthographic Sensitivity showed a significant main effect, $\beta = 0.677$, $SE = 0.318$, $t(57) = 2.13$, $p = .038$, $\eta_p^2 = .07$, consistent with the OS main effect reported in Table S2. Language Proficiency did not yield a reliable main effect, $\beta = 0.001$, $SE = 0.231$, $t(57) = 0.01$, $p = .996$. The interaction between Family Size and Orthographic Sensitivity was not significant, $\beta = 0.091$, $SE = 0.072$, $t(1618) = 1.27$, $p = .204$, indicating that, unlike Language

Proficiency, Orthographic Sensitivity did not modulate the N400 family size effect. The model explained 3.1% of variance via fixed effects ($R^2_{\text{marginal}} = .031$) and 75.6% overall ($R^2_{\text{conditional}} = .756$).

Table 6: Words N400, Family Size $\times$ OS (exploratory), fixed effects.

Term	β	SE	df	t	p
Family Size	0.452	0.064	1618	7.07	< .001
Orthographic Sensitivity	0.677	0.318	57	2.13	.038
Language Proficiency	0.001	0.231	57	0.01	.996
Family Size $\times$ OS	0.091	0.072	1618	1.27	.204

$R^2_{\text{marginal}} = .031$; $R^2_{\text{conditional}} = .756$. Random effects: participant, electrode nested within participant.

References